

Hu Zhiheng Henrey

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SUMMARY

Final-year Computer Science (Interactive Media & Game Development) student at Singapore Institute of Technology-DigiPen Institute of Technology joint degree programme. Learns by building, applying C++ and systems programming across multiple projects.

PROJECTS

Stretch

Year 2, 8 person team, C++

2D action platformer where players use a stretchy tongue to swing across gaps, climb ledges and defeat enemies.

- Built a JSON save and load system with RapidJSON covering about 30 data types, reusable templates and grid data, with categorised crash logging.
- Designed and solely wrote an undo/redo system for the team's in-house design tool, using JSON snapshots to reverse edits, creation and deletion.
- Developed a UI framework with 10 screens, buttons and sliders, supporting keyboard, mouse and gamepad input.
- Implemented behaviour logic as a 4 state machine in C++, after prototyping 4 behaviour types in Unity C#.
- Built an auto-save system that writes progress to a JSON file and restores the last saved state on failure.
- Served as AI & UI Champion on an 8 person team, owning enemy behaviour logic and the UI framework across 6 milestones.

WAVES

Year 1 team project, C++

Stealth maze game where players use wave-emitting rocks to see in the dark and evade enemies.

- Designed a state management system where every screen follows the same lifecycle, so screens switch cleanly with no memory leaks.
- Built the full user interface flow across 8 screens and overlays, including menus, settings, pause and results pages.
- Built a reusable UI component library with buttons, mouse hover detection and a font cache shared across all screens.
- Built a draggable volume slider linked to the app's global audio settings.
- Built the rendering system for the map's walls and boundaries.
- Led integration of four teammates' modules into one working application with end conditions.

EXPERIENCE

Private Tutor

2 years

- Tutored 4 students in Geography, Social Studies, Maths and Science, raising one student from a fail to B3 in Geography and Social Studies.
- Broke down difficult concepts and adapted lessons to each student's pace and progress, building my communication, patience and empathy.

Singapore Civil Defence Force (SCDF) — Emergency Medical Technician

Apr 2022 – Feb 2024

- Delivered emergency medical care as part of an ambulance crew over 22 months, making real-time triage and treatment decisions under pressure. Worked across a rotating team through high-stress emergency callouts, building the resilience to sustain performance under repeated pressure.

TongXin Technology — Industry Attachment

Dec 2019 – Jan 2020

- Observed engineers building an Arduino sensor-guided car and an OpenCV camera pipeline, gaining early exposure to embedded systems and computer vision.

EDUCATION

Singapore Institute of Technology (SIT)

Expected graduation 2028

BSc (Hons) Computer Science in Interactive Media and Game Development [Joint-Degree with DigiPen Institute of Technology Singapore]

Certifications: Google Data Analytics Professional Certificate (Coursera), 2024

SKILLS AND INTERESTS

Technical Skills: C++, C#, Python, JavaScript, SQL, HTML, Unity, Unreal Engine, Excel, Figma, OpenCV, Arduino, Git

Interests: Video/Photo Editing (Affinity, DaVinci Resolve), Dizi (Grade 9), Floorball (SCDF Inter-Division Champion, 2023)